How to Identify a Smoker?

An in-depth analysis on nation-wide behavior in relation to tobacco consumption

Smoking is a high-risk habit that puts all parts of the body at risk. Governments should be motivated in taking steps towards minimizing its country's tobacco consumption. In this research we set out to understand the environmental conditions that incentivize smoking behavior. We believe, that this approach could advance nationwide attempts at bettering the health of the people. Globally,

Our research looks at *tobacco consumption* and *the percentage of the population who are daily smokers* and uses them for a comparative analysis in relation to a variety of lifestyle measures. For a more meaningful perspective, we limited our focus on the countries with highest and lowest average percentage of daily smoker.

Regardless of the strong negative correlation suggested by the graphs, further research is needed both to further prove this phenomenon and to understand the potential underlying explanation of the relationship of smoking behavior and the weight of an individual.

While our findings are not definitive, we believe this approach should be further explored, introducing more measures and data.

why you decided on your analysis

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Data Science Project

The main measures used in the project:

These are all measures recorded between the years 1960-2023, either describing the share of population with that quality or an average, per capita measure.

Daily smokers: share of population who are daily smokers.

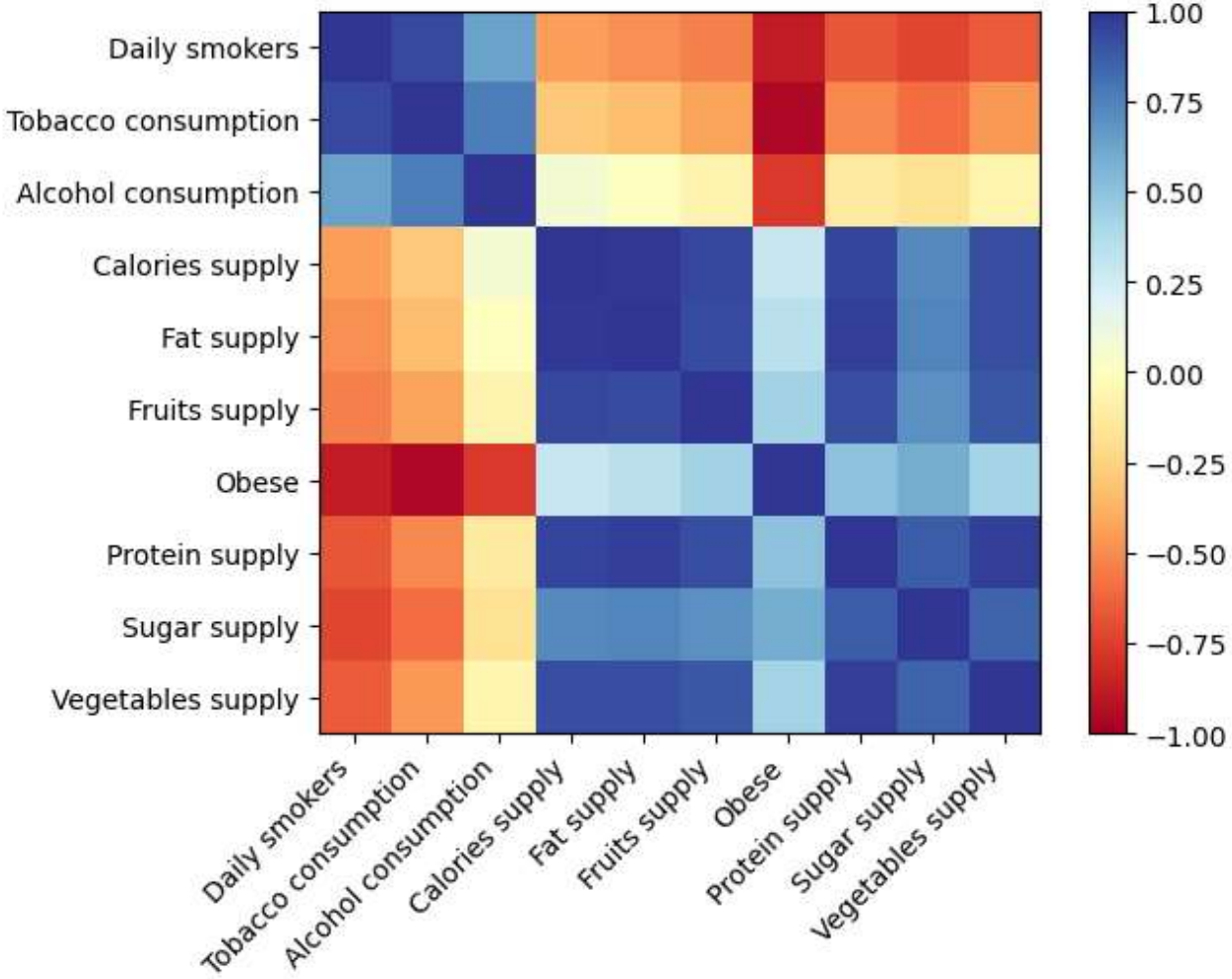
Obese: share of population who are obese.

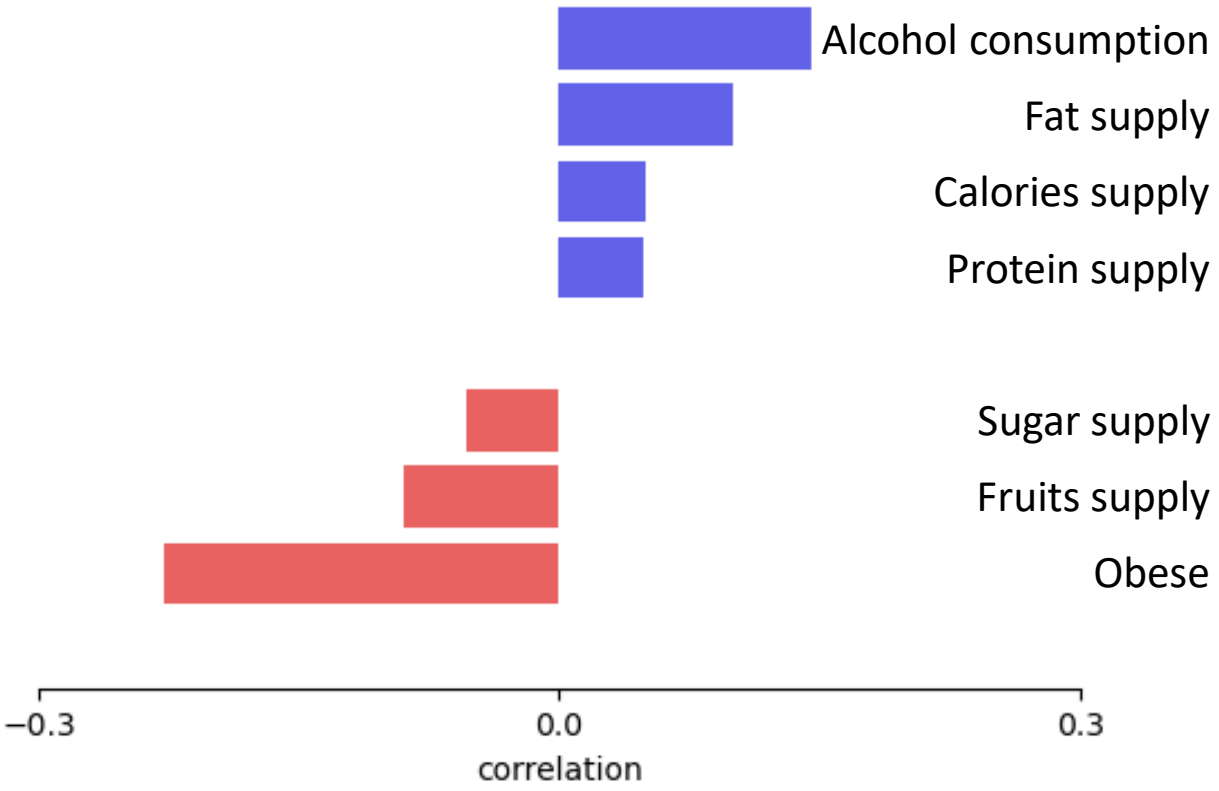
Tobacco, Alcohol consumption: on average per capita.

Calories, Fat, Fruits, Protein, Sugar, Vegetables supply: the amount available per capita.

Source: https://databank.worldbank.org/reports.aspx?source=2&series=SH.ALC.PCAP.FE.LI&country=

Correlation Matrix Between Measures
Across All Countries

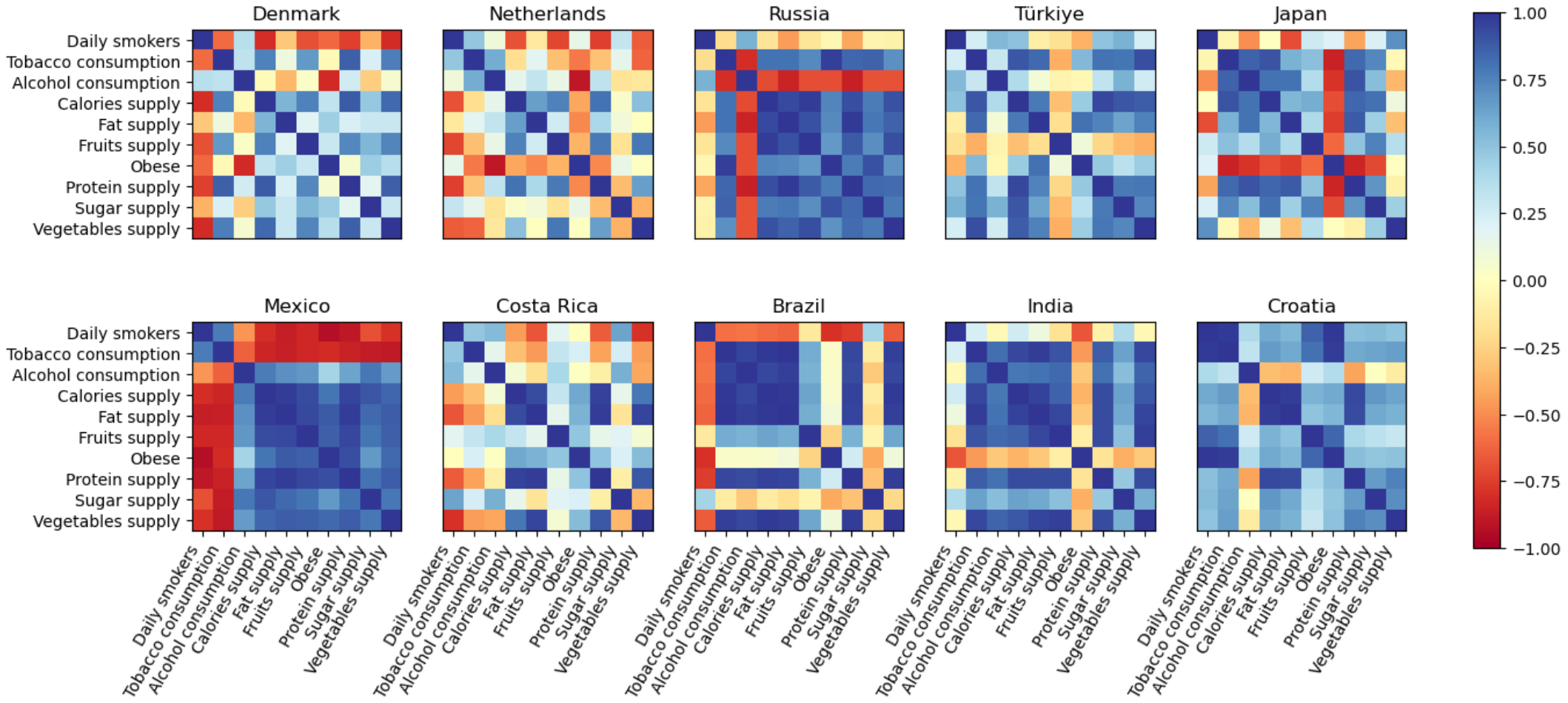




The plot shows a negative correlation globally between *Daily smokers, Tobacco consumption* and measures *Sugar, Fruits supply* and *Obese*.

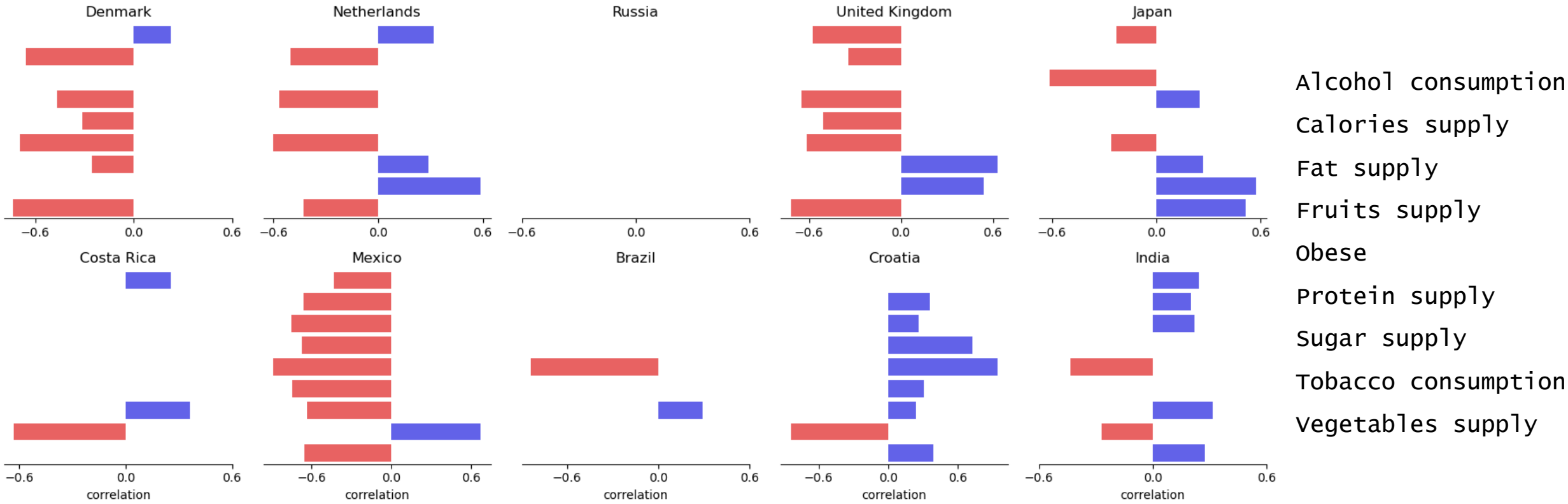
Correlation Matrices of Measures

In the Countries with the Highest (top row) and Lowest (bottom row) Average Percentage of Daily Smokers (PDS)



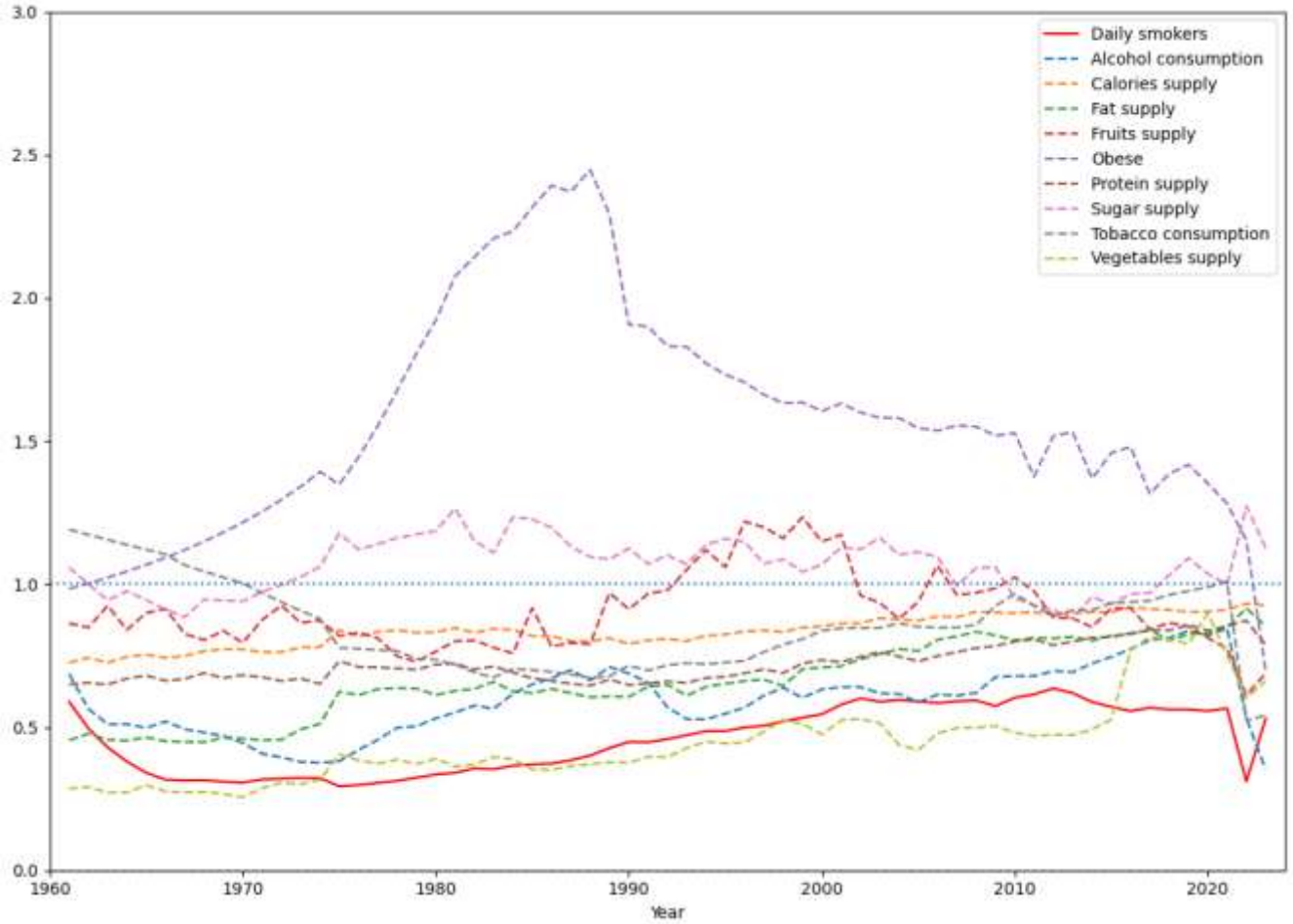
We do not see a clear pattern in the correlation matrices based on their *PDS*.

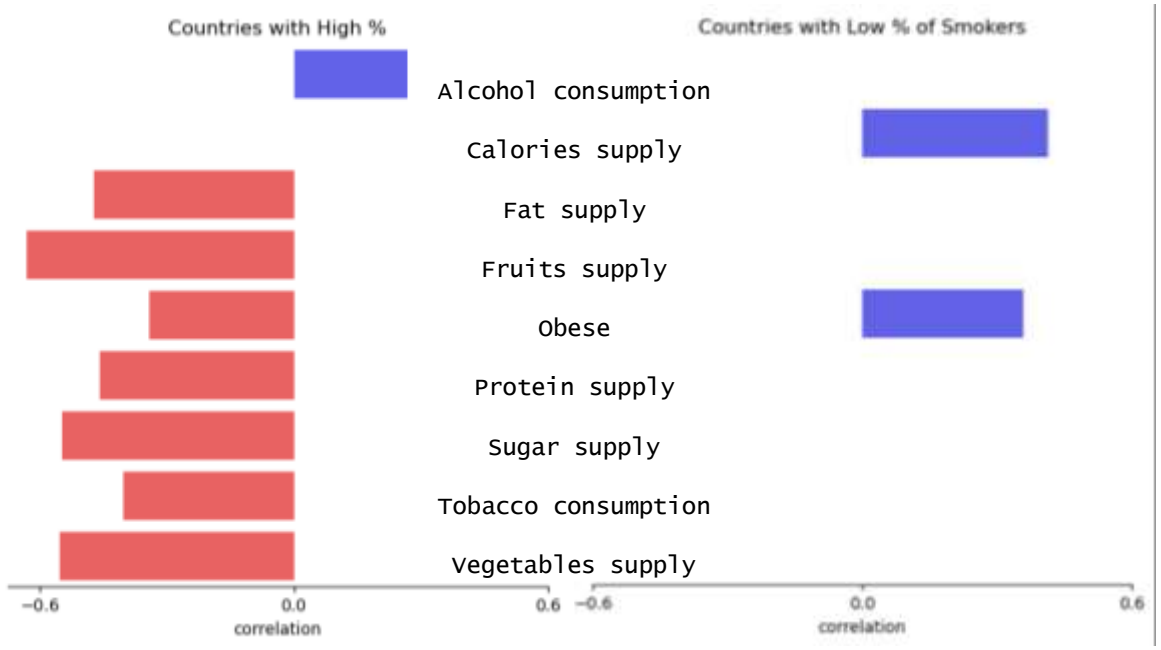
Correlation of *PDS* with other Features



Though we do not see a clear pattern in the correlation coefficients among our two group.

The Ratio of Measures in Countries with *low vs high PDS*





The measure *Obese* has a higher percentage in countries with a lower *PDS*.

This is consistent with our previous findings.